Geometric derivation of slerp formula

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Can you explain the derivation for the geometric formula for slerp?

Slerp (Spherical Linear Interpolation) is a beautiful piece of geometry. Let me walk through the derivation — it's essentially an exercise in decomposing one vector relative to another on a unit sphere.

The goal: find a smooth path between two unit vectors $\mathbf{p}$ and $\mathbf{q}$ on an n-sphere, parameterized by $t \in [0, 1]$, such that the arc length is traversed at constant angular velocity.

Setting up the geometry

Any point on the great-circle arc from $\mathbf{p}$ to $\mathbf{q}$ can be written as a linear combination of two orthogonal basis vectors. The natural choice is $\mathbf{p}$ itself, and a vector in the plane of $\mathbf{p}$ and $\mathbf{q}$ that is perpendicular to $\mathbf{p}$.

That perpendicular component is constructed by subtracting from $\mathbf{q}$ its projection onto $\mathbf{p}$:

$$\mathbf{q}_\perp = \mathbf{q} - (\mathbf{q} \cdot \mathbf{p})\,\mathbf{p}$$

Note that $\mathbf{q} \cdot \mathbf{p} = \cos\Omega$, where Ω is the angle between them. So:

$$\mathbf{q}_\perp = \mathbf{q} - \cos\Omega\,\mathbf{p}$$

The magnitude of this vector is $\sin\Omega$ (since $\mathbf{p}$ and $\mathbf{q}$ are unit vectors). Normalizing it:

$$\hat{\mathbf{q}}_\perp = \frac{\mathbf{q} - \cos\Omega\,\mathbf{p}}{\sin\Omega}$$

Parameterizing the arc

Now $\{\mathbf{p}, \hat{\mathbf{q}}_\perp\}$ form an orthonormal basis for the plane containing the arc. A point at angle $t\Omega$ from $\mathbf{p}$ along the arc is simply:

$$\mathrm{slerp}(\mathbf{p}, \mathbf{q}; t) = \cos(t\Omega)\,\mathbf{p} + \sin(t\Omega)\,\hat{\mathbf{q}}_\perp$$

Substituting the expression for $\hat{\mathbf{q}}_\perp$:

$$= \cos(t\Omega)\,\mathbf{p} + \sin(t\Omega) \cdot \frac{\mathbf{q} - \cos\Omega\,\mathbf{p}}{\sin\Omega}$$

Expanding and gathering the **p** terms:

$$= \left(\cos(t\Omega) - \frac{\sin(t\Omega)\cos\Omega}{\sin\Omega}\right)\mathbf{p} + \frac{\sin(t\Omega)}{\sin\Omega}\,\mathbf{q}$$

The coefficient of **p** simplifies via the sine subtraction identity sin(A − B) = sin A cos B − cos A sin B, with A = Ω and B = tΩ:

$$\frac{\sin(\Omega - t\Omega)}{\sin\Omega} = \frac{\sin((1-t)\Omega)}{\sin\Omega}$$

This gives the final formula:

$$\boxed{\mathrm{slerp}(\mathbf{p}, \mathbf{q}; t) = \frac{\sin((1-t)\Omega)}{\sin\Omega}\,\mathbf{p} + \frac{\sin(t\Omega)}{\sin\Omega}\,\mathbf{q}}$$

Here's a diagram of the geometric construction:

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Why this is "correct" interpolation

A few things worth noting about why this is the right formula:

- **Constant angular velocity**: the interpolated vector sweeps through equal angles per unit t, unlike lerp (plain linear interpolation) which speeds up near the midpoint and slows at the endpoints.
- **Stays on the sphere**: since it's written purely in terms of cos/sin of angles, the result always has unit magnitude.
- **Degeneracy at Ω ≈ 0**: when **p** and **q** are nearly parallel, sin Ω → 0 and the formula has a division-by-zero issue. In practice you fall back to normalized lerp (nlerp) in that regime, which is a good approximation for small angles.

- **$\Omega = \pi$** (antipodal points): the great-circle arc is not uniquely defined, so slerp is genuinely undefined there — you need to pick an intermediate point by convention.

The key insight of the whole derivation is just that **decomposing q into components parallel and perpendicular to p** gives you a clean orthonormal frame for the plane of rotation, and then you're just doing 2D polar interpolation.

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